

Homework 1

Please show all work for full credit. You're only allowed to use the standard built-in Python 3 libraries.

Question 1. Watch [Mary's Room: A philosophical thought experiment – Eleanor Nelsen](#) on YouTube.

Do you think there is something about human learning that is different than machine learning?

Yes

Question 2. Rate your skills out of ten in the following subjects,

- 1) Probability and Statistics
- 2) Calculus
- 3) Linear Algebra
- 4) Python

- 1) 1
- 2) 1
- 3) 1
- 4) 3

Question 3. Of the sub-plots (a), (b), (c) and (d) in figure , state which ones have no correlation, negative correlation, position correlation and non-linear correlation.

- (a) position
- (b) negative
- (c) no correlation
- (d) non-linear

Question 4. Assume that the probability of Alice going to class everyday given she got a good grade is $\frac{5}{6}$ and the probability of her getting a good grade is $\frac{2}{5}$ while the probability of her going to class everyday is $\frac{1}{3}$. What is the probability that Alice gets a good grade given she went to class everyday?

Hint: Thomas Bayes.

$$\frac{5}{6} * \frac{2}{5} / \frac{1}{3} = 1$$

Question 5. We define the natural numbers as,

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$$

Let x be the arithmetic mean of the first $2^{33} - 1$ natural numbers. What is $\log_2(x)$?

Hint: Carl Friedrich Gauss.

$$\log(2^{32}) = 32$$

Question 6. Let `arr = [x % 2 for x in range((2**100)-1)]` be a Python list using [Python list comprehension](#). What is the statistical mode of the list `arr`? Justify your answer.

0 since odd ranges have an extra 0

Question 7. Define a function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(x) = \frac{\sin(x^2)}{y}$$

Calculate the following,

- The partial derivative with respect to y , $\frac{\partial f}{\partial y}$
- The partial derivative with respect to x , $\frac{\partial f}{\partial x}$
- The gradient vector $\nabla f(x, y)$.

- 1) $-y^{-2} \sin(x^2)$

- 2) $\frac{2\cos(x)x}{y}$
 3) $[\frac{2\cos(x)x}{y}, -y^{-2}\sin(x^2)]$

Question 8. Let $\vec{v}_1 = [e, \pi, \sqrt{2}]$ and $\vec{v}_2 = [1, 2, 0]$. What is the dot product $\vec{v}_1 \cdot \vec{v}_2$? Please do not give an approximation.

$$(e)(1) + (\pi)(2) + 0 = e + 2\pi$$

Question 9. We define a matrix of n rows and p columns with real values as $\mathbf{A} \in \mathbb{R}^{n \times p}$. We can think of $\mathbf{A}$ as having n row vectors with the $i^{\text{th}} \leq n$ row vector being $\text{row}(\mathbf{A})_i$. Similarly, we can think of $\mathbf{A}$ as having p column vectors with the $j^{\text{th}} \leq p$ column vector being $\text{col}(\mathbf{X})_j$.

Then matrix multiplication between $\mathbf{A} \in \mathbb{R}^{n \times p}$ and $\mathbf{B} \in \mathbb{R}^{p \times m}$ is defined as,

$$\mathbf{AB} = \begin{bmatrix} \text{row}(\mathbf{A})_1 \text{col}(\mathbf{X})_1 & \text{row}(\mathbf{A})_1 \text{col}(\mathbf{X})_2 & \cdots & \text{row}(\mathbf{A})_1 \text{col}(\mathbf{X})_m \\ \text{row}(\mathbf{A})_2 \text{col}(\mathbf{X})_1 & \text{row}(\mathbf{A})_2 \text{col}(\mathbf{X})_2 & \cdots & \text{row}(\mathbf{A})_2 \text{col}(\mathbf{X})_m \\ \vdots & \vdots & \ddots & \vdots \\ \text{row}(\mathbf{A})_n \text{col}(\mathbf{X})_1 & \text{row}(\mathbf{A})_n \text{col}(\mathbf{X})_2 & \cdots & \text{row}(\mathbf{A})_n \text{col}(\mathbf{X})_m \end{bmatrix}$$

where $\text{row}(\mathbf{A})_i \text{col}(\mathbf{X})_j \in \mathbb{R}$ is a dot or inner product.

Further, the transpose of $\mathbf{A} \in \mathbb{R}^{n \times p}$ is written as $\mathbf{A}^T \in \mathbb{R}^{p \times n}$ and is defined as,

$$\text{If } \mathbf{A}^T \text{ is the transpose of } \mathbf{A} \text{ then } \text{row}(\mathbf{A})_i = \text{col}(\mathbf{A}^T)_i.$$

- (a) What must be true of the number of columns of $\mathbf{A}$ and number of rows of $\mathbf{B}$ for $\mathbf{AB}$ to be defined?

The number of cols in A must match number of rows in B

- (b) What information do the number of rows of $\mathbf{A}$ and number of columns of $\mathbf{B}$ give you about the dimensions of the product $\mathbf{AB}$?

AB will have the same amount of rows as A and the same amount of columns as B has

- (c) For the two matrices bellow, give the product $\mathbf{AB}$.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 4 & 2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 4 & 3 \\ 8 & 9 & 5 \\ 6 & 5 & 3 \\ 11 & 9 & 6 \end{bmatrix}$$

- (d) Prove that the matrix product is not a symmetric relation, i. e., $\mathbf{AB} \neq \mathbf{BA}$. Hint: can you construct a small counter example? In 9.c, AB is a valid operation, but BA is not since B does not have as many rows as A has columns

- (e) Prove that $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$.

$$(\mathbf{AB})^T = \begin{bmatrix} 5 & 8 & 6 & 11 \\ 4 & 9 & 5 & 9 \\ 3 & 5 & 3 & 6 \end{bmatrix} \quad \mathbf{B}^T \mathbf{A}^T = \begin{bmatrix} 5 & 8 & 6 & 11 \\ 4 & 9 & 5 & 9 \\ 3 & 5 & 3 & 6 \end{bmatrix}$$

- (f) For $\mathbf{X} \in \mathbb{R}^{n \times p}, \beta \in \mathbb{R}^p, y \in \mathbb{R}^n$ prove that,

$$\beta^T \mathbf{X}^T y = y^T \mathbf{X} \beta$$

If we assume $n = 1$ and $p = 1$, then $\beta^T = \beta, X^T = X, y^T = y$ then $\beta^T \mathbf{X}^T y = y^T \mathbf{X} \beta$ could be written as $\beta X y = y X \beta$ which is true

- (g) Let $f(\beta) = \beta^T \beta$. Prove that the derivative $\frac{df}{d\beta} = 2\beta$.

if we assume $\beta = [x, x]$ then $\beta^T \beta = x^2 + x^2 \frac{df}{d\beta} = 2x + 2x$ which is equivalent to 2β

Question 10. For $x \in \mathbb{R}$ we have,

$$f(x) = \frac{x^4}{4} - \frac{x^3}{3}$$

Give,

$$\min_x f(x) = \min_x \left(\frac{x^4}{4} - \frac{x^3}{3} \right)$$

Plot f to double check your answer.

$x = 1$

Question 11. Let $x \sim \mathcal{N}(0, 1)$ be a normally distributed random variable with mean 0 and standard deviation 1. What is the likelihood that $x = \sqrt{2}$. Give an exact answer.

Question 12. The code snippet in listing reads the plain text of the novel *The Cosmic Computer* by the famous American science fiction writer Henry Beam Piper over the internet and saves it to a string variable `text`.

```

1  from urllib.request import urlopen as get
2
3  url = 'https://www.gutenberg.org/files/20727/20727.txt'
4  with get(url) as response:
5      text = response.read().decode('utf-8')
6
7  print(text)

```

LISTING 1. A Python program to download the text of the *The Cosmic Computer* by Piper.

Write a Python program that prints out the ten most used words in the novel that have more than 5 letters. State your findings. Put your code in a file called `frequency.py`.

Merlin 129 father 108 around 79 didn't 77 Maxwell 71 anything 66 Project 64 people 62 things 61 something 60

Question 13. Watch [Laziness in Python - Computerphile](#) on YouTube.

Implement the function `fibonacci(n)` in the code listing

```

1  def fibonacci(n):
2      # implement me
3
4  for t in fibonacci(50):
5      print(t)

```

LISTING 2. A Python 3 program to print out the first $n \geq 0$ Fibonacci numbers.

Run your program for $n = 50$. What are the last five numbers printed? Put your code in a file called `fibonacci.py`.

1134903170 1836311903 2971215073 4807526976 7778742049

SUBMISSION INSTRUCTIONS

- 1) Submit a PDF that answers all the questions.
- 2) Submit Python files, e. g., `frequency.py` and `fibonacci.py`.

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